

Vision-based mmWave Beam Prediction in V2X Communications using JEPA

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Abstract—This work presents the first application of Joint Embedding Predictive Architecture (JEPA) to mmWave beam prediction in Vehicle-to-Everything (V2X) communications. The model learns image representations from DeepSense 6G, a large-scale V2X dataset, using self-supervised masked visual prediction. Three adaptation strategies are evaluated for beam prediction using the learned representations. They are frozen encoder weights, Low-Rank Adaptation (LoRA) fine-tuning, and fine-tuning the last few layers of the transformer. The frozen encoder performs poorly on the beam prediction task. However, the LoRA fine-tuning significantly improves the accuracy, reaching about 95% for top-3 accuracy on one of the beam prediction tasks. Binary classification tasks for day/night and LOS/NLOS both also exceed 96% accuracy with frozen encoder weights, demonstrating good learned representations for simpler tasks. Evaluation on an out-of-distribution dataset also shows strong cross-dataset generalization capabilities of the model. The results show some promise for developing foundational models for V2X communication systems.

Index Terms —Vehicle-to-Everything (V2X), Millimeter-wave (mmWave), Joint Embedding Predictive Architecture, Low-Rank Adaptation (LoRA), Beam prediction

I. INTRODUCTION

Vehicle-to-Everything (V2X) supports reliable, low-latency communication for autonomous driving. However, mmWave has limited penetration and is easily blocked, causing the link to degrade quickly. Beam selection is critical in ensuring stable mmWave links in V2X. Traditional beam selection relies on beam sweeping or hierarchical codebook search to find the best beam. The challenge with these techniques is that the associated control overhead and delay are often incompatible with short contact times and quick scene changes in vehicular environments [1]. There has been a shift towards multimodal machine learning (MML) for beam selection. This involves using multiple sensors such as camera, LiDAR, radar and GPS instead of a single modality. This gives the vehicle a richer context of the environment so the best beam can be selected. Existing studies have shown that fusing data from multiple modalities leads to more accurate mmWave beam predictions compared to using one a single modality or relying on exhaustive beam sweeping [2]–[6]. These multimodal models, however, have some limitations. For instance, they are trained in a supervised manner so they require large amounts of labeled data for each new deployment environment. They are also optimized for specific datasets so they do not generalize when deployed in new or unseen scenarios. MML models are also built for just one task and that is beam selection or prediction. They cannot perform other tasks without retraining from scratch. These limitations impede the scalability and

practical deployment in real-world scenarios where multiple tasks must be done simultaneously.

In recent times, foundational models have become a popular alternative to supervised learning models. These models are a form of self-supervised learning (SSL) that learn generalizable representations from large amounts of unlabeled data, eliminating the need for expensive labeled datasets. Once pretrained, a single foundational model can perform multiple downstream tasks by training only a lightweight head on a small amount of labeled data. This makes them more scalable than conventional machine learning (ML) methods. In the wireless communication domain, foundational models use input modalities such as raw In-phase and Quadrature (IQ) samples and spectrograms to perform downstream tasks like RF signal classification and beam prediction [7], [8]. In mmWave beam prediction for V2X communication, foundational models present a promising direction because they can learn generalizable representations from unlabeled vehicular sensing data (LiDAR, radar, images) and be adapted to different downstream tasks. The application of self-supervised learning to mmWave beam prediction remains limited and no existing work has explored JEPA-based foundational models. This gap motivates the need for a more scalable approach to beam prediction in V2X systems.

SSL approaches for wireless communications that rely on contrastive learning learn representations by pulling similar pairs closer together and pushing dissimilar pairs apart in the embedding space. However, this requires carefully crafted data augmentations and positive-negative sample pairs that are often domain-specific. Contrastive learning can also suffer from false negatives which leads to slow convergence and loss of semantic information. This motivates moving toward prediction-based, non-contrastive approaches such as Joint-Embedding Predictive Architectures (JEPA). JEPA-based models learn by predicting the latent representations of masked input regions from the original image without requiring any augmentations [9]. This allows the model to capture more meaningful and generalizable representations from unlabeled data.

This work proposes a JEPA-based foundational model for mmWave beam prediction in V2X communication systems. A Vision Transformer encoder is pretrained on unlabeled vehicular images from the DeepSense 6G [10]. The pretrained encoder is then evaluated on three downstream tasks, that is, beam prediction, day or night classification and line-of-sight (LOS) and non-line-of-sight (NLOS) classification. The main contributions of this work are:

- A JEPA-based foundational model is pretrained on unlabeled vehicular camera images from the DeepSense 6G dataset, representing one of the first applications of JEPA-based mmWave beam prediction in V2X communications.
- A comprehensive analysis of the pretrained encoder across three downstream tasks: beam prediction, LOS/NLOS classification, and day/night scene classification. The results demonstrate the foundational model's ability to perform different downstream tasks from the learned representation.
- Few-shot adaptation on the e-FLASH dataset show that the pretrained encoder successfully adapts to an out-of-distribution environment with minimal labeled data. This demonstrates that the learned representations capture generalizable visual features that extend beyond the training distribution.

II. RELATED WORKS

ML has emerged as a more efficient alternative to beam sweeping for selecting the most optimal beam in mmWave V2X communications. Many works have proposed ML-based beam prediction frameworks using different sensing modalities such as camera (images), LiDAR, GPS and radar. For vision-based mmWave prediction, authors in [11] propose a vision-aided beam tracking system using images captured on a camera as the input. They use an encoder-decoder architecture which is evaluated on the real-world DeepSense 6G dataset. Vision-aided beam tracking on mmWave bands, combining camera images and historical beam indices to predict future optimal beams is also investigated in [12]. A deep learning approach using an autoencoder for image feature extraction and a bidirectional GRU for sequential modeling is proposed and evaluated on a reformulated version of the ViWi dataset [13]. Similarly in [14], sequence-to-sequence (Seq2seq) deep learning is proposed for mmWave beam prediction using images and historical beam indices. CNN-based feature extraction and beam index embedding are combined with a GRU-based Seq2seq model and attention mechanism to predict future beam directions.

Recently, MML has seen increased adoption in beam prediction due to its superior performance over single-modality approaches. In [2], an MML framework that integrates images and positional data to enable accurate beam prediction for a vehicle is proposed. Features extracted from visual data using a CNN are concatenated with positional (longitude and latitude) inputs processed by an MLP to predict the optimal beams. Similarly in [3], the authors propose a fusion-based deep learning approach for mmWave beam selection for vehicular applications using GPS, images and LiDAR. The framework fuses latent embeddings from each modality to predict the top-K candidate beam pairs, narrowing the beam search space. Multi-modal learning (MML) is also explored in [4], where data from GPS, camera, radar, and LiDAR are jointly fused, and the resulting feature vectors are processed using an LSTM network to utilize historical observations for beam prediction. Large Language Models (LLMs) have also been explored

for mmWave beam prediction. In [15], the authors propose M²BeamLLM, a framework that combines images, radar, LiDAR and GPS with a fine-tuned GPT-2 model to reformulate beam prediction as a multimodal reasoning task. Supervised fine-tuning is used to adapt the LLM to the beam prediction task.

The previous works discussed mostly rely on task-specific supervised models, each requiring its own architecture or model that cannot perform any other task apart from beam prediction. This limits their generalization because a separate model must be designed and trained for each new task or scenario. This motivates the adoption of foundational models, where rich and transferable representations learned from unlabeled data can be efficiently adapted to diverse downstream wireless communication tasks.

III. METHODOLOGY

This section describes the dataset used, the JEPA pretraining framework, and the approach for adapting the pretrained encoder to the downstream tasks.

A. Dataset

For this work, the dataset used is the DeepSense 6G dataset [10]. The DeepSense6G dataset is a large-scale real-world multi-modal dataset designed for sensing-aided wireless communication research. It consists of synchronized multi-modal data collected across diverse outdoor and indoor environments, including LiDAR, radar, RGB camera images, GPS and mmWave beam measurements. The RGB camera, a ZED2 from StereoLabs, operates at 30 frames per second (fps) with a 110° field of view (FOV). The mmWave communication system operates at 60 GHz, where the transmitter (mobile user) uses a quasi-omnidirectional antenna to radiate signals, and the receiver (base station) consists of a 16-element phased array with a 60 GHz RF front-end. The receiver performs beam sweeping using a 64-beam codebook over a 90° FOV to measure received signal power which serve as ground truth for the beam prediction tasks. For this work, RGB images are the modality used since JEPA is a vision-based self-supervised framework. Pretraining is performed on scenarios 1-16, 24 and 30-35 which makes up approximately 112,000 images.

B. I-JEPA pretraining

The pretraining framework adopted in this work is the Image-based Joint Embedding Predictive Architecture (I-JEPA), a self-supervised learning method that learns rich visual representations by predicting the representations of masked image regions in latent space instead of reconstructing pixels [9]. Given an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times C}$, it is first divided into a sequence of non-overlapping patches which are projected into a sequence of patch tokens fed into an encoder.

I-JEPA has three main components. It uses two encoders, that is, a context encoder E_θ and a target encoder $E_{\bar{\theta}}$ and a lightweight predictor network P_ϕ . A multi-block masking strategy is adopted, where a set of M target blocks $\mathcal{T} = \{t_1, t_2, \dots, t_M\}$ is sampled from the image as the masked

regions the model must predict, while a non-overlapping context block is provided as the visible region for the context encoder. The context encoder, E_θ , processes the visible patches and produces context representations while the target encoder, $E_{\bar{\theta}}$, processes the full image and produces the target patch representation. The predictor, P_ϕ , takes the context representations along with the positional information of each target block and predicts the corresponding target representations. The pretraining objective minimizes the mean squared error (MSE) between the predicted and actual target representations:

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|\hat{z}_t - z_t\|_2^2 \quad (1)$$

where $\hat{z}_t$ is the predicted representation and z_t is the target representation produced by $E_{\bar{\theta}}$ for target block t , and $|\mathcal{T}|$ denotes the total number of target blocks. The context encoder E_θ is updated via gradient descent, while the target encoder $E_{\bar{\theta}}$ is updated as an exponential moving average (EMA) of E_θ :

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta \quad (2)$$

where $\tau \in [0, 1]$ is the EMA momentum coefficient, ensuring the target encoder provides stable and consistent representations throughout training. Fig. 1 shows the pretraining architecture with the various components.

The I-JEPA pretraining is done in the Advanced Research Computing (ARC) cluster using the Nvidia H100 GPU. The parameters used for the pretraining task are summarized in Table I.

TABLE I: I-JEPA Pretraining Parameters

Parameter	Value
Encoder	ViT-Base
Image Size	224×224
Patch Size	16×16
Epochs	100
Batch Size	256
Learning Rate	6.0×10^{-4}
Predictor Embedding Dim	768
Weight Decay	0.04
Start Learning Rate	2.0×10^{-4}
Final Learning Rate	1.0×10^{-6}
Optimizer	AdamW
Number of Target Blocks	4

C. Downstream Tasks

For downstream tasks, $E_{\bar{\theta}}$ is used for feature extraction instead of E_θ . This is because E_θ only processes partial masked images during pretraining, whereas $E_{\bar{\theta}}$ always receives the full unmasked image so it produces complete representations. Also, since $E_{\bar{\theta}}$ is updated via EMA rather than direct gradient descent, its weights are more stable and produce smoother representations. Thus, this makes it better suited for downstream evaluation.

During downstream evaluation, $E_{\bar{\theta}}$ is frozen and mean pooling is applied over the patch token representations from the last transformer block to produce a fixed-size feature vector. A lightweight linear layer or MLP head is attached on top of the extracted features depending on the task, and

trained in a supervised manner using only a small portion of labeled data. The three downstream tasks considered in this work are beam prediction, day/night classification, and LOS/NLOS classification. Lastly, an out-of-distribution task is considered, where the pretrained encoder is evaluated on a completely different dataset without any retraining. This is done to assess the ability of the learned representations to generalize beyond the pretraining domain.

IV. EXPERIMENTS AND RESULTS

This section presents the experimental setup, results, and analysis for each downstream task. Downstream experiments are conducted in the ARC cluster and Google Colab, depending on the task. Performance is evaluated using top- k accuracy for beam prediction and standard classification accuracy for the day/night and LOS/NLOS tasks. Top- k accuracy measures whether the optimal beam index appears within the top- k predicted beams. It is defined as:

$$\text{Top-}k \text{ Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} [y_i \in \hat{y}_i^{(k)}] \quad (3)$$

where N is the total number of test samples, y_i is the ground truth beam index for sample i , $\hat{y}_i^{(k)}$ is the set of top- k predicted beam indices, and $\mathbf{1}[\cdot]$ is the indicator function. This metric is relevant for beam prediction because having the optimal beam within the top- k predictions is sufficient for maintaining reliable communication in practice. The results for each of the downstream tasks are compared to a ResNet-18 baseline.

A. Beam Prediction

For the beam prediction task, scenarios 5 and 9 were randomly chosen. Each scenario has a corresponding CSV file that contains detailed metadata for each image including, GPS coordinates, image filename, timestamps and best beam index. The ground-truth best beam indices are gotten from the CSV files and used as targets for the prediction task. A multi-layer perceptron (MLP) was used as the prediction head to capture non-linear relationships between the learned visual representations and the optimal beam indices. Three adaptation strategies were explored to see their impact on beam prediction accuracies. The techniques are (i) frozen encoder weights with the MLP head trained, (ii) Low-Rank Adaptation (LoRA) fine-tuning of the encoder and (iii) unfreezing the last few layers of the encoder for fine-tuning. Results are reported for $k \in \{1, 2, 3, 5\}$

1) *Frozen Encoder Weights*: As discussed earlier, the weights of the encoder are frozen and an MLP task head is attached for the beam prediction.

Table II shows the performance of the ResNet-18 and the frozen I-JEPA encoder on scenario 5 and 9. The frozen encoder performs poorly relative to the ResNet-18 baseline. For instance, the frozen encoder's top-5% accuracy is only 71.1% for scenario 9 while the ResNet-18 baseline achieves a top-5% performance of 71.7% for the same scenario. Generally, both models perform better in scenario 5 than 9. The poor performance could be attributed to the mismatch between the

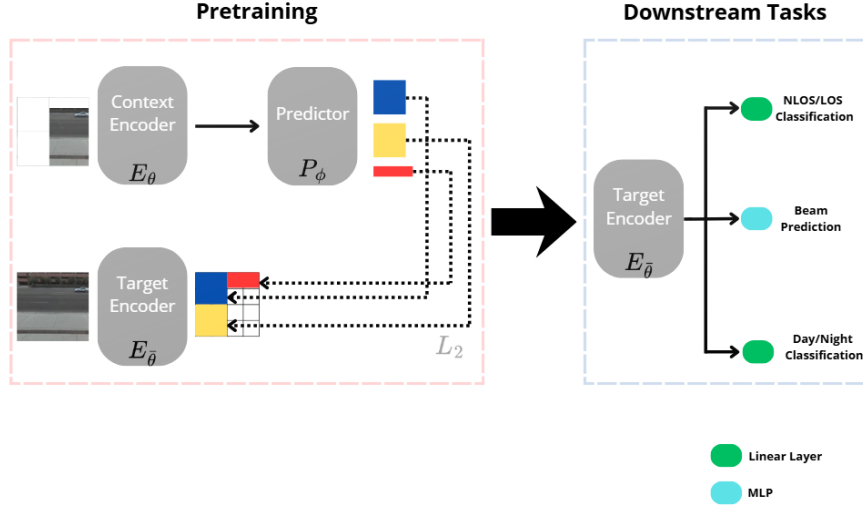


Fig. 1: I-JEPA architecture with pretraining and downstream tasks

TABLE II: Top-k beam prediction accuracy for ResNet-18 and I-JEPA

Scenario	Model	Top-1 (%)	Top-2 (%)	Top-3 (%)	Top-5 (%)
5	ResNet-18	32.2	56.6	69.3	81.8
	I-JEPA	20.1	38.2	52.7	73.1
9	ResNet-18	54.0	79.6	91.7	97.8
	I-JEPA	23.7	41.7	58.9	71.7

TABLE III: Top-k beam prediction accuracy for ResNet-18 and LoRA fine-tuned I-JEPA

Scenario	Model	Top-1 (%)	Top-2 (%)	Top-3 (%)	Top-5 (%)
5	ResNet-18	32.2	56.6	69.3	81.8
	I-JEPA	36.5	59.6	71.1	83.4
9	ResNet-18	54.9	79.6	91.7	97.8
	I-JEPA	60.2	85.1	94.9	98.6

pretraining objective and the downstream task. However, to further understand the poor performance, a confusion matrix was obtained for scenario 5 for the frozen encoder. Fig. 2 shows the confusion matrix for the top-20 most frequent beams in scenario 9. The confusion matrix reveals that misclassifications predominantly occur between adjacent beam indices rather than random beams. For instance, beam 2 is correctly classified 50% of the time but confused with beam 3 (45%). Misclassifications are concentrated around neighboring beams. This indicates that the model learns the general angular region of the best beam but struggles to differentiate between closely-spaced beam directions.

2) *LoRA fine-tuning*: LoRA is a parameter-efficient fine-tuning technique that freezes the pretrained model weights and injects trainable low-rank matrices into each layer. Instead of updating all model parameters, LoRA introduces small adapter matrices that learn task-specific modifications while preserving the original pretrained knowledge. The rank parameter (r) determines the dimensionality of these adapter matrices, controlling their capacity, while alpha (α) acts as a scaling factor that amplifies the adapter's influence on the final output.

While running the test, it was observed that increasing both α and r increased the accuracy significantly. However, this improvement came at the expense of longer training times due to the increased number of trainable parameters. After experimenting with different values, the selected α and r values selected were 32.

Table III shows the top-k beam prediction accuracy for ResNet-18 and LoRA fine-tuned I-JEPA across scenarios 5

and 9. The LoRA-adapted I-JEPA consistently outperforms the ResNet-18 baseline across all top-k metrics in both scenarios. LoRA improves the performance by adapting the pre-trained representations for beam prediction while preserving the model's learned knowledge.

3) *Unfreezing the last N-layers*: This technique involves freezing most of the encoder while allowing the final N transformer blocks to update their weights during the downstream tasks. This provides more adaptation flexibility than the frozen encoder but requires lower learning rates to preserve pretrained knowledge. In this work, the last 4 transformer blocks were unfrozen to enable them learn task-specific features while the earlier layers retained the general visual representations from pretraining.

TABLE IV: Top-k beam prediction accuracy for ResNet-18 and fine-tuning the last 4 blocks

Scenario	Model	Top-1 (%)	Top-2 (%)	Top-3 (%)	Top-5 (%)
5	ResNet-18	32.2	56.6	69.3	81.8
	I-JEPA	20.1	39.2	54.0	73.6
9	ResNet-18	54.9	79.6	91.7	97.8
	I-JEPA	23.5	44.0	59.5	79.2

Fine-tuning the last 4 transformer blocks slightly improves the accuracy for all scenarios. This technique allows the final layers to adapt to beam-specific patterns while earlier layers remain frozen. However, this approach requires updating many parameters compared to LoRA fine-tuning. Thus, it is com-

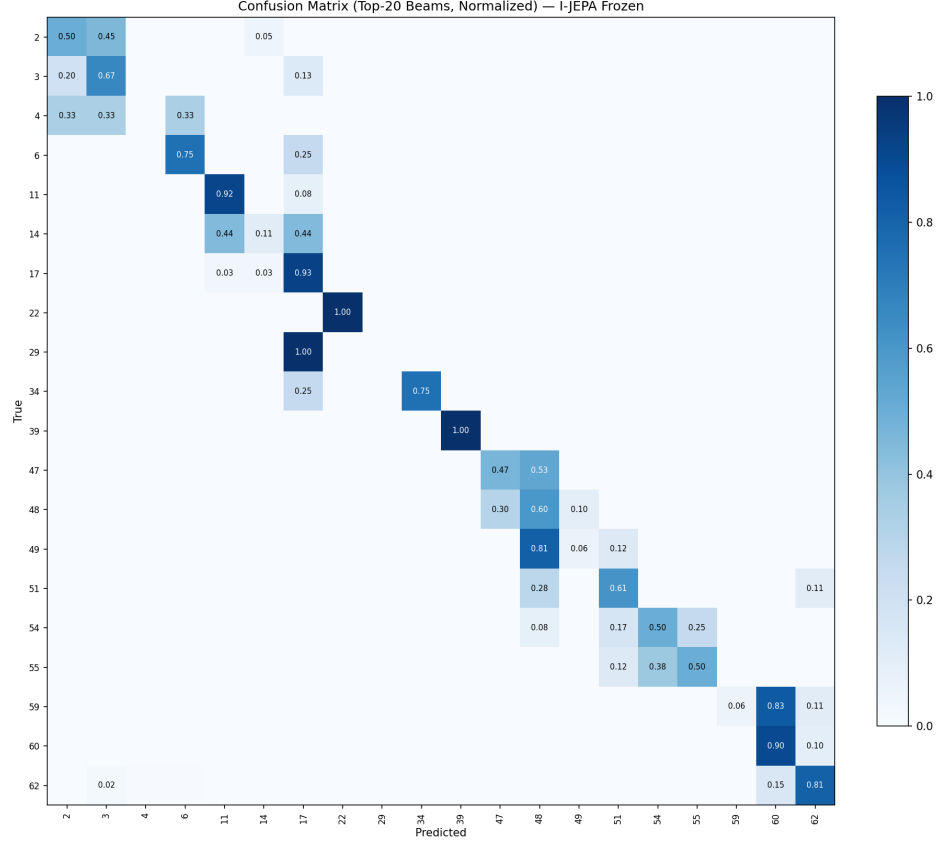


Fig. 2: Confusion matrix for beam prediction on scenario 9 showing the normalized prediction distribution across all beam classes.

putationally expensive (has more trainable parameters) while achieving a lower performance compared to LoRA.

direct line between them, it is classified as LOS as shown in Fig. 3. In the dataset, LOS is labeled as 0 and NLOS as 1.

B. NLOS/LOS Prediction

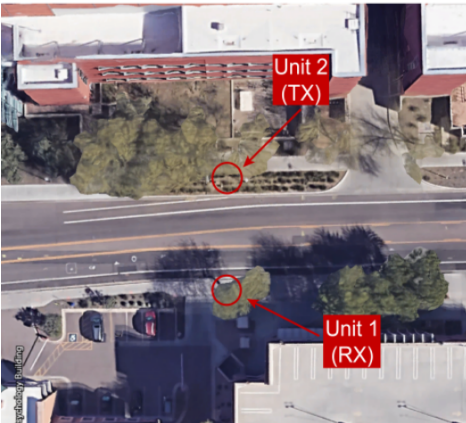


Fig. 3: LOS/NLOS scenario setup

For the LOS/NLOS classification task, scenarios 24 and 30 were used. In this setup, the transmitter and receiver are positioned on opposite sides of a street. When a vehicle obstructs the direct path between the transmitter and receiver, it is labeled as NLOS. Conversely, when there is an unobstructed

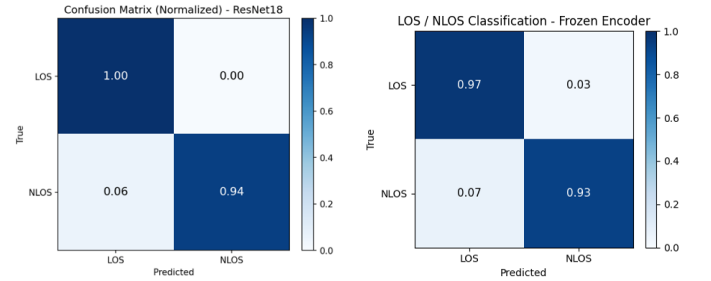


Fig. 4: Confusion matrix for NLOS/LOS.

Fig. 4 shows the confusion matrix of the NLOS/LOS prediction between the ResNet-18 and the frozen I-JEPA encoder. Both models achieve strong performance with ResNet-18 getting **97.7%** accuracy and the frozen I-JEPA encoder achieving **97.2%** accuracy. ResNet-18 perfectly classifies all LOS samples (100% recall) while correctly identifying 94% of NLOS cases. Both models show a pattern of higher false negative rates for NLOS samples compared to LOS samples. This indicates that vehicle blockage presents more visual ambiguity than clear LOS conditions. The 0.5 percentage point difference demonstrates that pretrained representations transfer effectively to LOS/NLOS classification without requiring full

fine-tuning of the model.

C. Day/Night Prediction

The data recordings for all scenarios were taken either during the day or in night. For this task, 6 scenarios for each class were selected. Scenarios were labeled based on recording conditions i.e. day scenarios (1, 3, 6, 7, 8, 9) assigned label 0 and night scenarios (2, 4, 5, 14, 33, 34) assigned label 1.

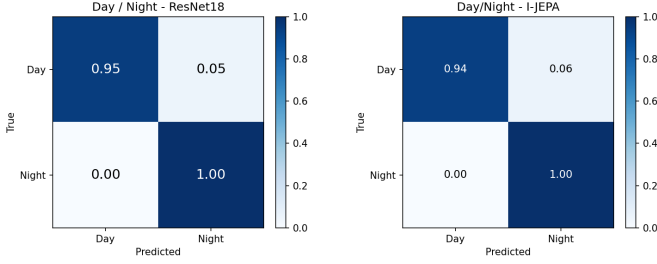


Fig. 5: Confusion matrix for Day/Night

Fig. 5 shows the confusion matrix of the day/night classification task. The ResNet-18 has an accuracy of **96.4%** while the frozen I-JEPA encoder gets an accuracy score of **96.2%**. Both models never misclassify night images as day. However for day images, there is some misclassifications. A plausible reason for this is that night images are easier to distinguish due to their consistently distinctive low-light characteristics, whereas daytime images captured during overcast or twilight conditions can exhibit reduced brightness that many resemble nighttime lighting. The similar performance between the two models shows that I-JEPA effectively learns lighting-related features during pretraining.

D. Out-of-Distribution Performance

To evaluate the generalization capability of the pretrained representations to out-of-distribution data, few-shot adaptation experiments were conducted using the e-FLASH dataset [16]. The e-FLASH dataset is a real-world multimodal V2X dataset containing synchronized camera images, LiDAR point clouds, and GPS data collected from an autonomous vehicle in urban environments. The goal was to assess how well the encoder, pretrained on DeepSense 6G, could adapt to e-FLASH with a few labeled examples. For evaluation, an episode labeled as NLOS (Category 2, left to right, episode 3) in the E-Flash dataset was selected because NLOS conditions present a more challenging beam prediction scenario. A ResNet-18 was used as the baseline model for comparison.

TABLE V: Few-shot adaptation I-JEPA vs ResNet-18 on e-FLASH Category 2 Episode 3

Shots	Model	Top-1 (%)	Top-2 (%)	Top-3 (%)	Top-5 (%)
10	ResNet-18	91.9	98.8	100	100
10	I-JEPA	88.4	98.8	100	100
20	ResNet-18	98.0	100	100	100
20	I-JEPA	94.0	100	100	100
30	ResNet-18	96.6	100	100	100
30	I-JEPA	96.6	100	100	100

Table V shows the results for the few-shot adaptation. Both models perform well on this dataset and achieve 100% accuracy for the top-3 and top-5 for all number of shots. This is significantly better than their performance on DeepSense 6G scenarios. A likely explanation for this performance is that the e-FLASH dataset uses a controlled data collection environment with limited spatial and temporal variability, as all samples were captured on a single street during daytime conditions. In contrast, DeepSense 6G contains diverse vehicular scenes with changing traffic, buildings, and lighting conditions. Thus, beam prediction task is easier in the e-Flash dataset. This validates that I-JEPA generalizes well to new scenarios. However, domain-specific pretraining would provide greater advantages in more complex and diverse environments.

V. CHALLENGES

The main challenge faced was getting a high score for the accuracy of the beam prediction task. For the frozen encoder, the accuracy was much lower than the ResNet-18 baseline. This was, however, improved by using LoRA fine-tuning although the training process was much slower. Images alone cannot give enough context about an environment for beam selection, especially when there are complex 3D spatial relationships between the transmitter, receiver, and obstacles that cannot be fully captured from a single 2D image perspective. There are also factors like signal reflections, multi-path effects and material properties that influence RF propagation but cannot be captured visually. Consequently, the relationship between scene appearance and the optimal beam was often indirect and difficult for the model to learn.

Other challenges faced include the organization of the dataset which required significant reorganization before training could begin. The images were originally stored across multiple scenario-specific folders which required consolidation into a unified directory structure for efficient data loading. Also, transferring over 50 GB of dataset from Google Drive to the ARC cluster was time-consuming and required careful handling to ensure all files were correctly copied.

VI. CONCLUSION & FUTURE WORKS

This work investigated foundation models for mmWave beam prediction and binary classification in V2X communication. The I-JEPA encoder was pretrained on the DeepSense 6G dataset using masked patch prediction. It performed well on binary classification tasks, achieving over 96% accuracy in both day/night and LOS/NLOS classification. These results show that the pretraining successfully captured visual features relevant for understanding V2X environments. Beam prediction, however, was more challenging. The frozen encoder performed poorly on this task. LoRA fine-tuning, however, significantly improved performance, demonstrating that pretrained representations can effectively support geometric reasoning tasks when given task-specific adaptation. Three adaptation strategies were compared i.e. frozen encoder, LoRA fine-tuning, and unfreezing the last 4 layers. LoRA fine-tuning had the best accuracy. The pretrained encoder also demonstrated strong generalization to the e-FLASH dataset,

an out-of-distribution V2X dataset with different environmental conditions and sensor configurations. Few-shot adaptation experiments showed that the model could effectively transfer learned representations.

For future work, there are many promising directions. CNN-JEPA and other foundational models could be explored to assess their performance on the beam prediction task. Different masking strategies could also be explored to see how that impacts the performance of the downstream task. Also, incorporating other modalities such as LiDAR and radar data alongside camera images would add depth and material information that vision alone cannot capture. This multimodal fusion could significantly improve beam prediction accuracy. More complex downstream tasks, such as object detection, would better evaluate the versatility of pretrained V2X representations. To the best of my knowledge, this work represents the first application of I-JEPA to beam prediction tasks. It paves the way for future exploration of foundation models in V2X communication systems.

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